



GTC4Lusso USA - Removing the complete PTU - Latest Update: 2021-01-26

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C4.02 Removing the complete PTU

Removing the complete PTU



The workshop tests performed after installing a new component must include tests of all safety system components. If the "Safety NDA" cycle performed with the Deis tester is failed, the PTU is NOT in safe mode or the PTU registers any errors, do NOT return the vehicle to the customer.



In the event of a Pcode with the additional code TSR, contact the SAT Help Desk.



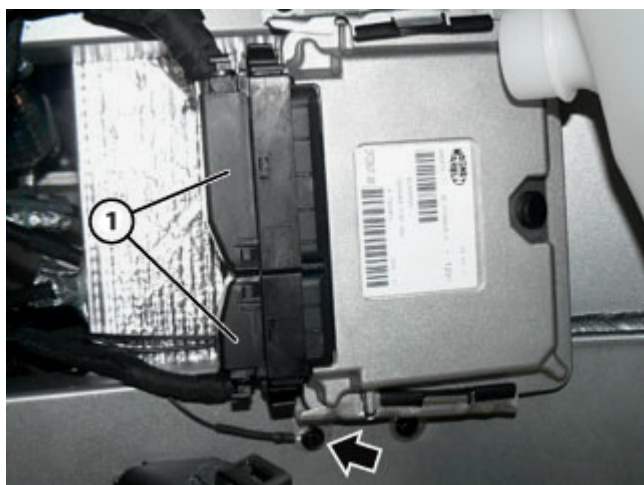
In case of disassembly prior to shipment, the oil levels in the ATF and GL circuits before draining must be specified.

Any warranty claims must also include a photograph showing the oil level in the ATF reservoir before the circuit is drained as proof.

- Disconnect the battery ( F2.01).
- Remove the engine compartment cross-member ( E2.04).
- Remove the front bumper ( E3.03).
- Remove the steering box ( D5.04).
- Remove the front stabiliser bar ( D4.05).
- Remove the front axle shafts ( C5.01).
- Detach the front axle shafts from the PTU, leaving them connected to the hub carriers.
- Drain the PTU gear oil system ( C4.06).
- Remove the front wheelhouses ( E3.05).



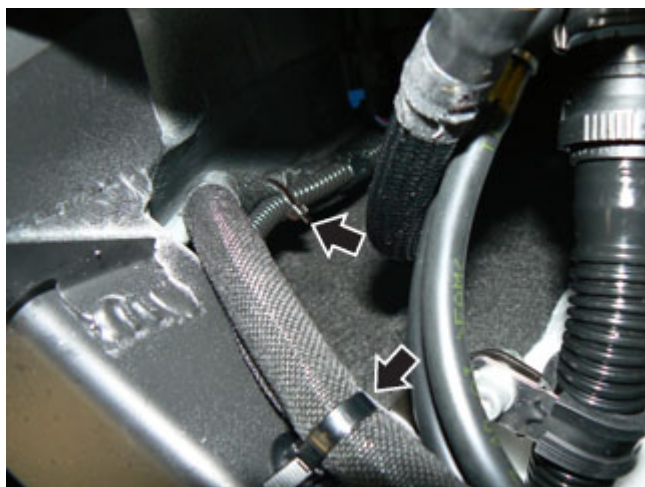
- Disconnect the ground cables, loosening the indicated screws.



- Disconnect the ground cable by loosening the indicated screw.
- Disconnect the connectors **(1)**.



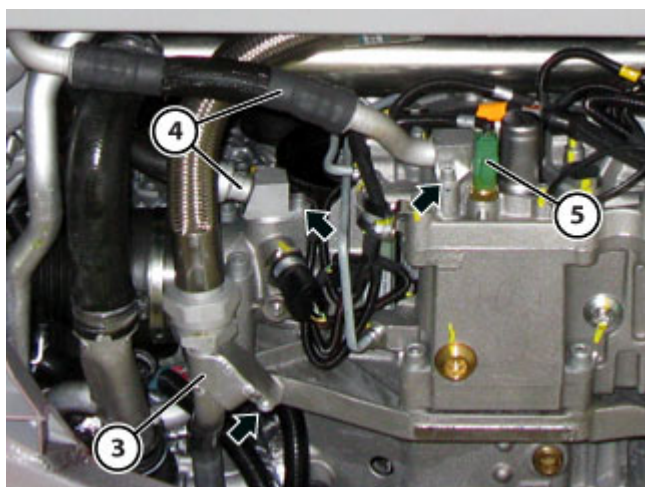
- Disconnect the connector **(2)** and release the respective cable, detaching the indicated clamps.



- Detach the cable by releasing the indicated clips.



- Detach the cable by undoing the indicated nut fastening the rubber-lined bracket.

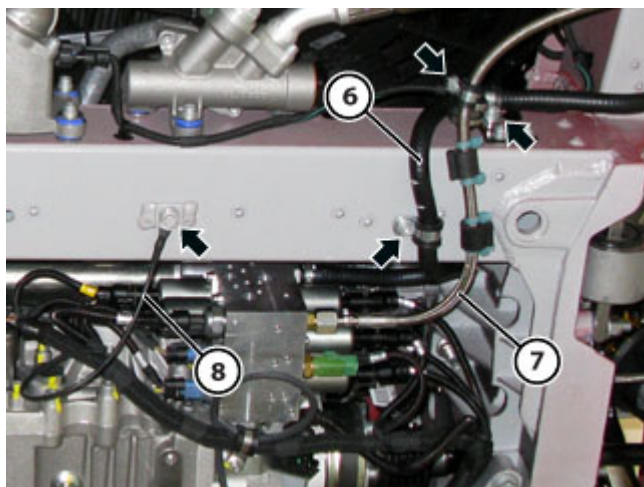


- Undo the indicated screw fastening the engine oil pipe **(3)**.

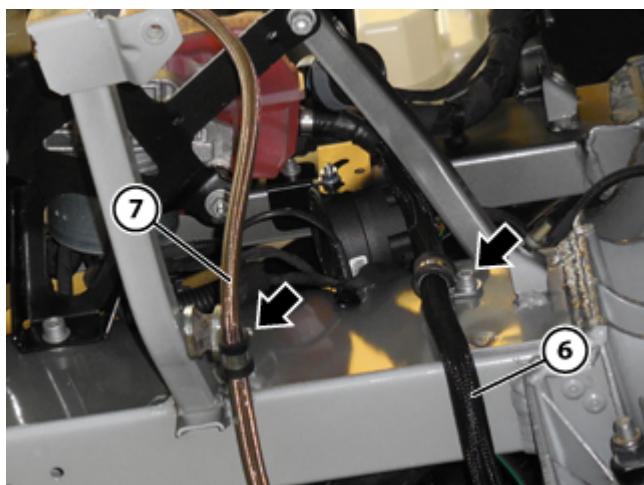


ALWAYS dispose of liquids correctly in accordance with applicable legislation. Collect all liquids drained from the vehicle systems in appropriate airtight containers, then close the containers.

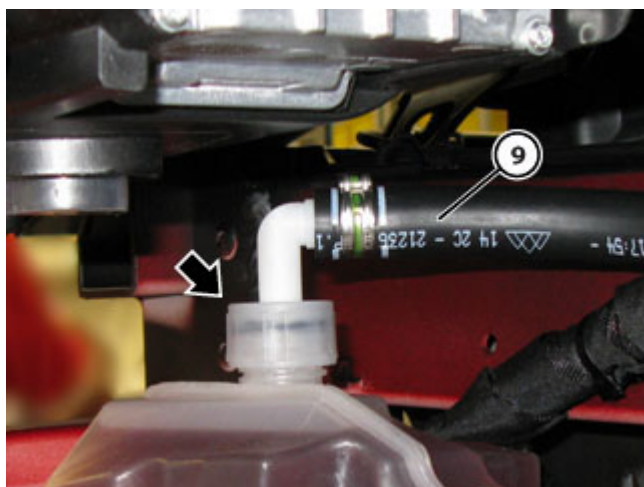
- Place suitable recipients for collecting the oil under the pipe unions **(4)**.
- Disconnect the oil pipes **(4)**, loosening the respective screws indicated.
- Disconnect the connector **(5)**.



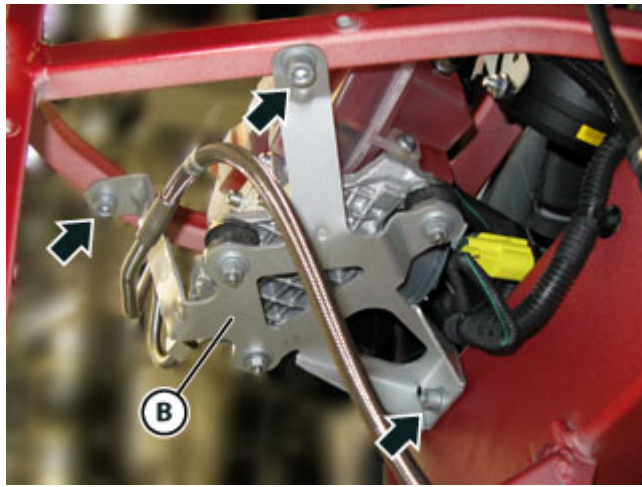
- Release the PTU system electrohydraulic actuator system pipe **(6)** by undoing the indicated screws.
- Release the PTU system electrohydraulic actuator system pipe **(7)** by undoing the indicated nut.
- Disconnect the ground cable **(8)** by undoing the indicated screw.



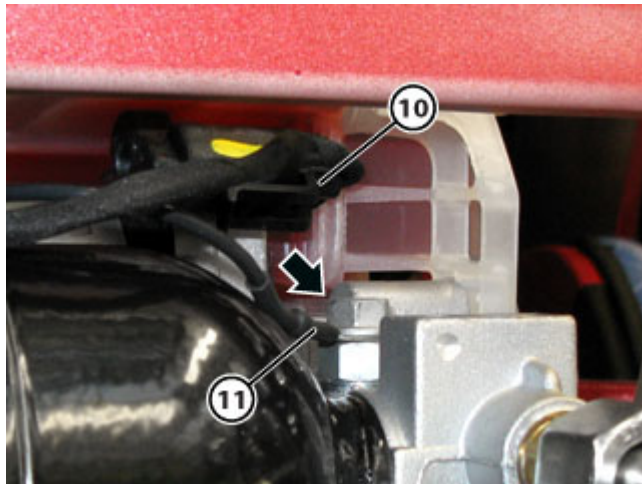
- Release the PTU system electrohydraulic actuator system pipe **(6)** by undoing the indicated screw.
- Release the PTU system electrohydraulic actuator system pipe **(7)** by undoing the indicated nut.



- Disconnect the pipe **(9)**, undoing the indicated threaded ring.
- Plug the pipe **(9)** and the respective connection on the PTU pneumatic actuator fluid tank.

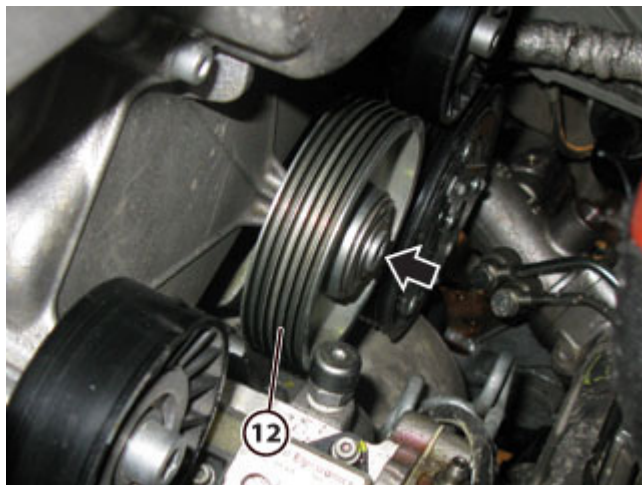


- Remove the bracket **(B)**, complete with the PTU pneumatic actuator oil reservoir.



- Disconnect the connector **(10)**.
- Disconnect the ground cable **(11)** by undoing the indicated screw.

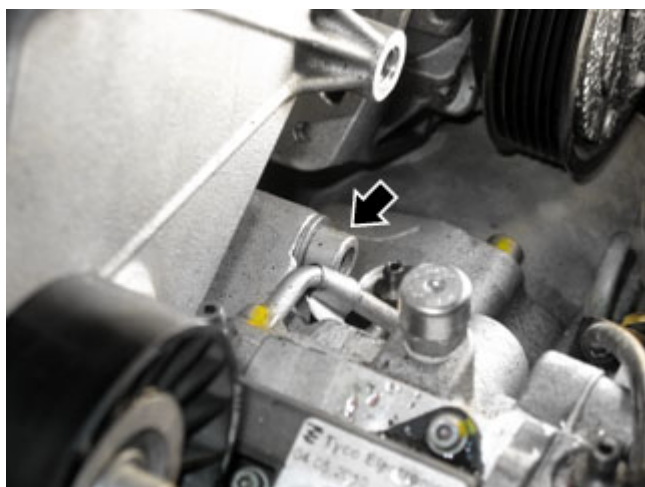
- Note the respective positions of each of the screws fastening the PTU to the engine, as the screws are of differing length.



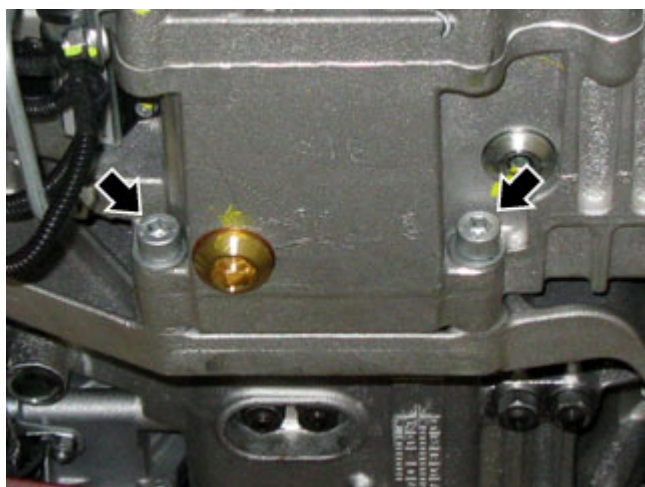
- Remove the pulley **(12)**, undoing the indicated screw.



- Undo the indicated nut.



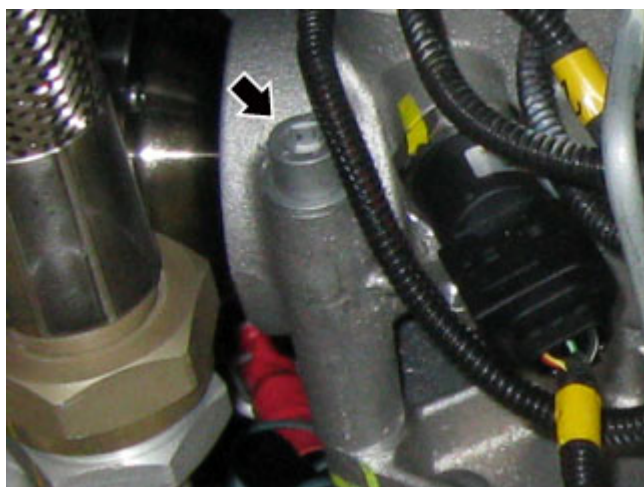
- Undo the indicated screw.



- Undo the indicated screws.



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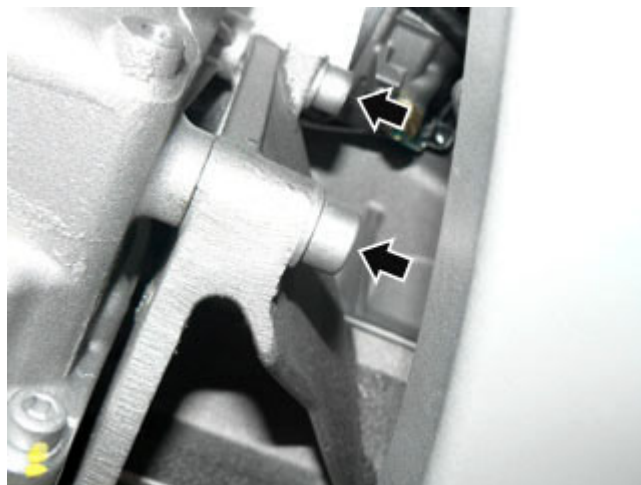
- Connect the tool (**AV 8353**) to a suitable hydraulic jack.
- Manoeuvre the hydraulic jack to position the tool (**AV 8353**) under the PTU, with the lateral support bracket on the left.
- If necessary, lower the lower pads with the respective knobs (**13**).



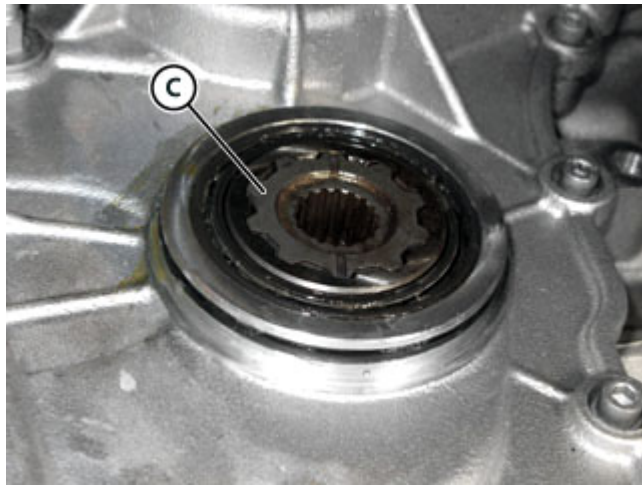
- Align the lateral support bracket of the tool (**AV 8353**) with the front LH axle shaft flange, to match the respective fastener holes.
- Tighten the indicated screws.



- Manoeuvre the hydraulic jack to position the tool (**AV 8353**) under the PTU, with the lateral support bracket on the left.
- Lift the lower pads of the tool (**AV 8353**) using the respective knobs (**13**), to bring them against the PTU.



- Undo the indicated screws.



- Using the hydraulic jack and the tool (**AV 8353**), separate the PTU from the engine, taking care to keep the decoupling shaft (**C**) in its relative seat on the engine.



- Taking care not to knock any surrounding components, lower the complete PTU (**D**) out of the engine compartment and set down in a safe place